

智能结构与智慧港口

Smart Marine Metastructures

浙江大学海洋学院
Ocean college, Zhejiang University

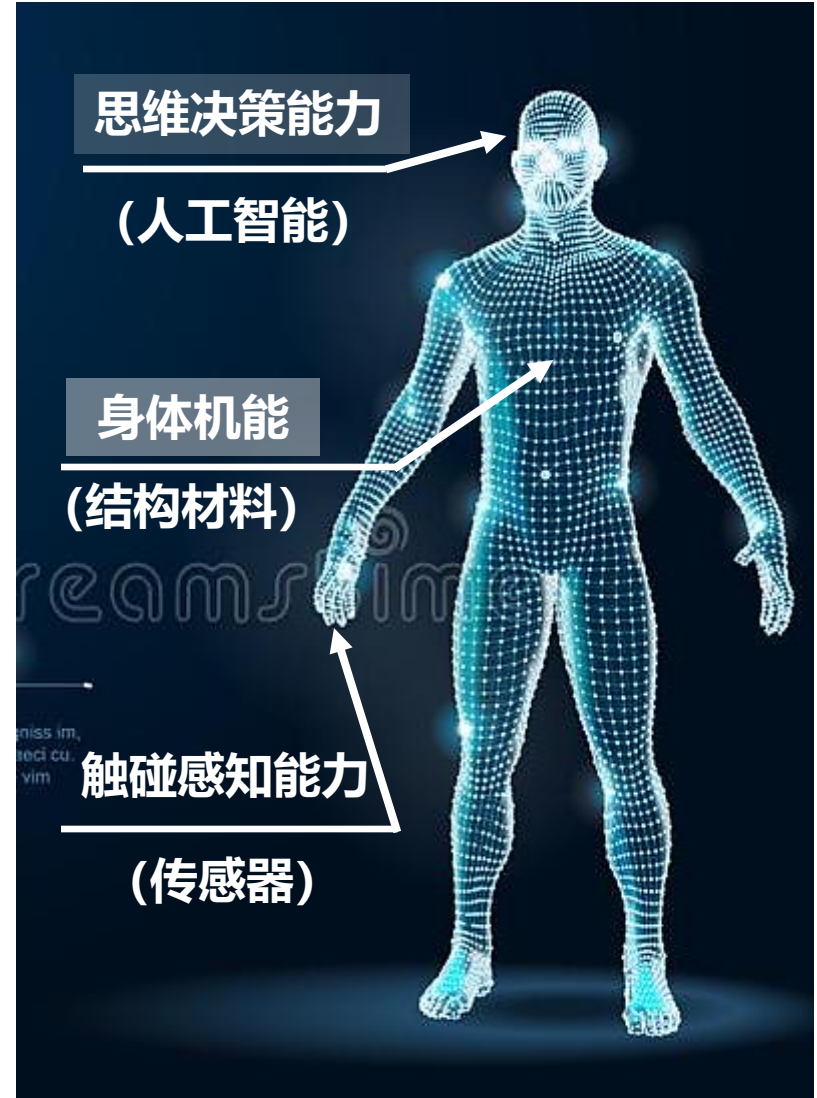


CALENDERS

Week	Subject	Content
1	Introduction and Overview	1. Overview of structural and material analysis 2. Introduction of structural materials and vision for their applications in ocean engineering
2	Advanced Metastructures	1. Structures vs. metastructures 2. Advanced characteristics of metastructures
3	Structural Health Monitoring (SHM)	1. Advent and development of structural health monitoring (SHM) 2. Main SHM technologies
4	SHM in Civil and Ocean Engineering	1. SHM in civil and ocean engineering 2. Challenges and future work
5	Introduction to Intelligent Ports	1. Traditional ports vs. modern intelligent ports 2. Information networks of intelligent ports
6	Equipment and Cases of Intelligent Ports	1. Typical equipment in intelligent ports 2. Case study of intelligent ports
7	Structural Design and SHM in Data Era	1. Vision for smart marine metastructures and intelligent ports 2. Summary of the class
8	Oral Presentation	

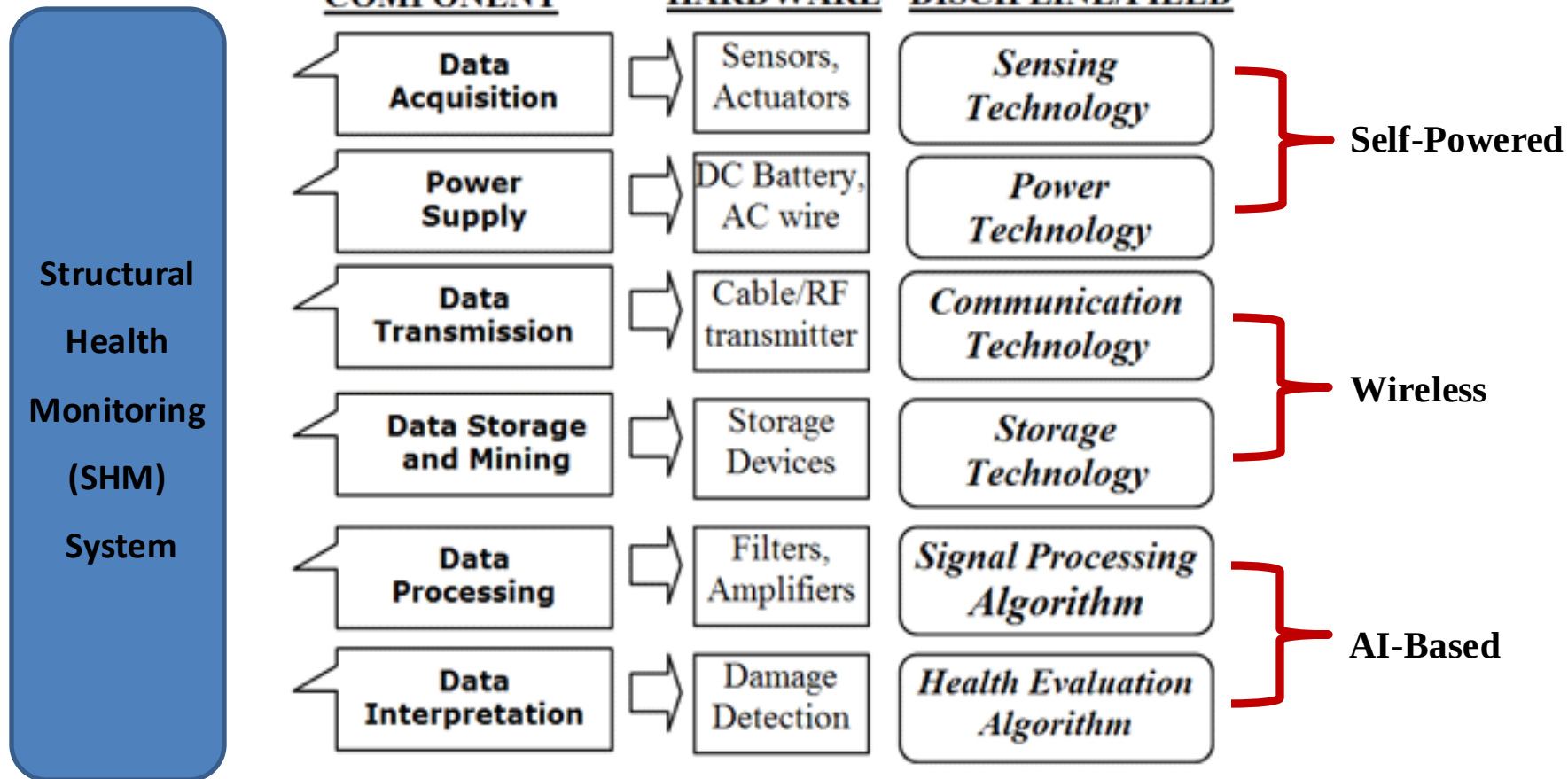
《智能结构与智慧港口》

- ❑ What is intelligence?
- ❑ Why is intelligence of interest?
- ❑ How to be intelligent?
- ❑ When is intelligence achievable?



Structural Health Monitoring (SHM)

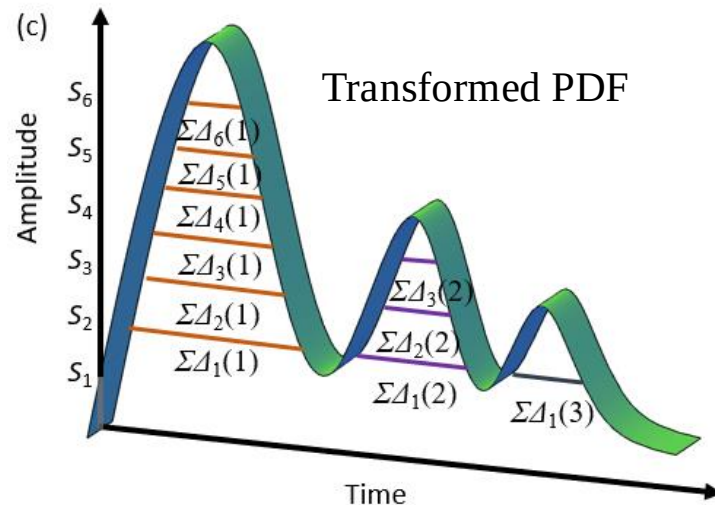
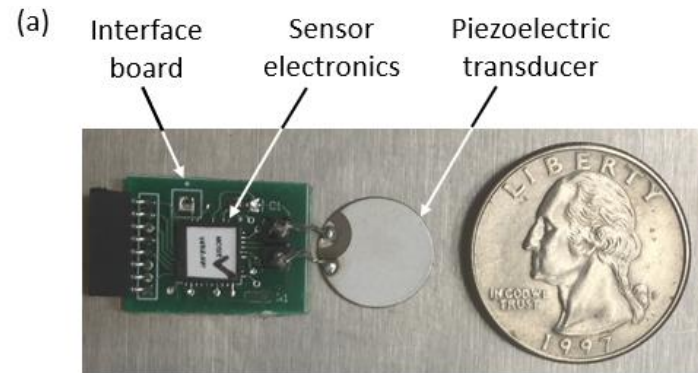
Background



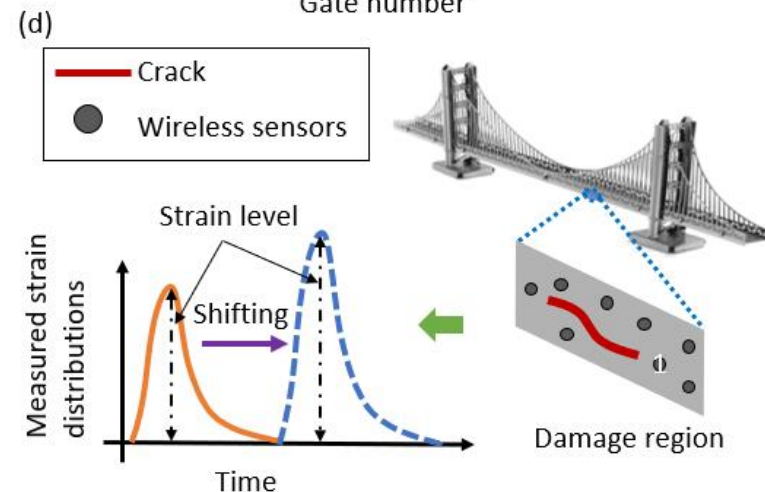
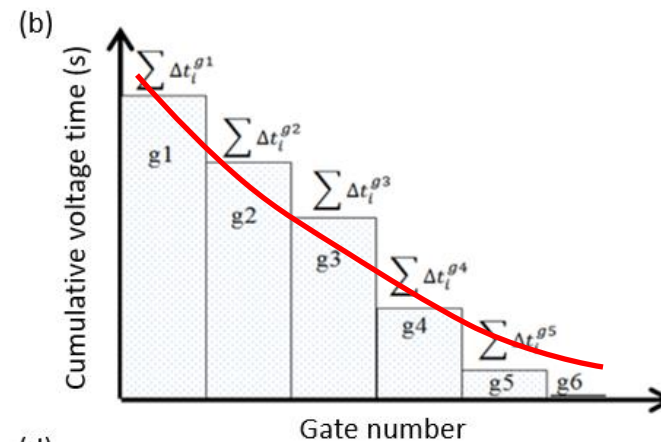
Structural Health Monitoring (SHM)

Self-powered, Wireless Sensors

Initial Detection

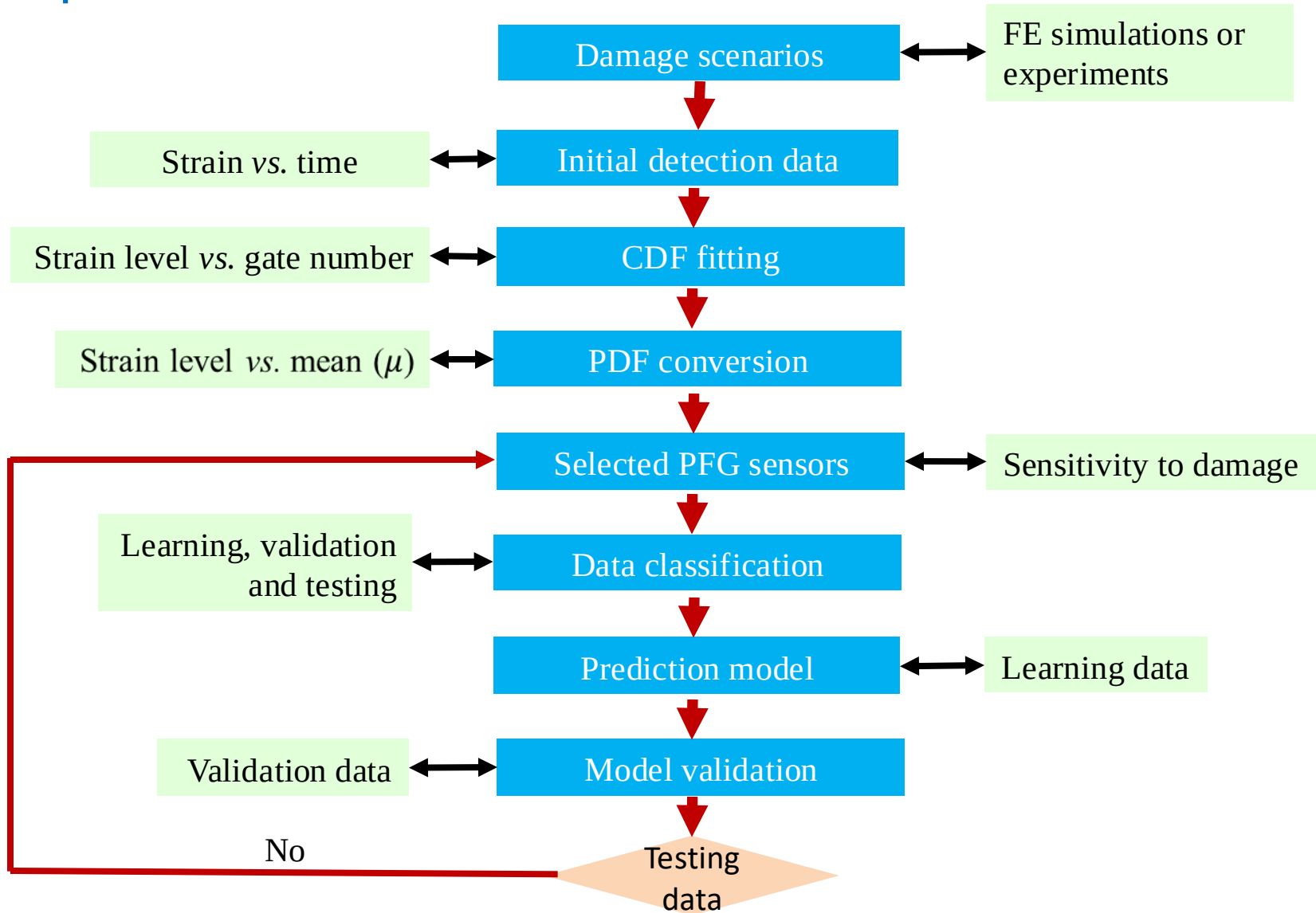


CDF Fitting



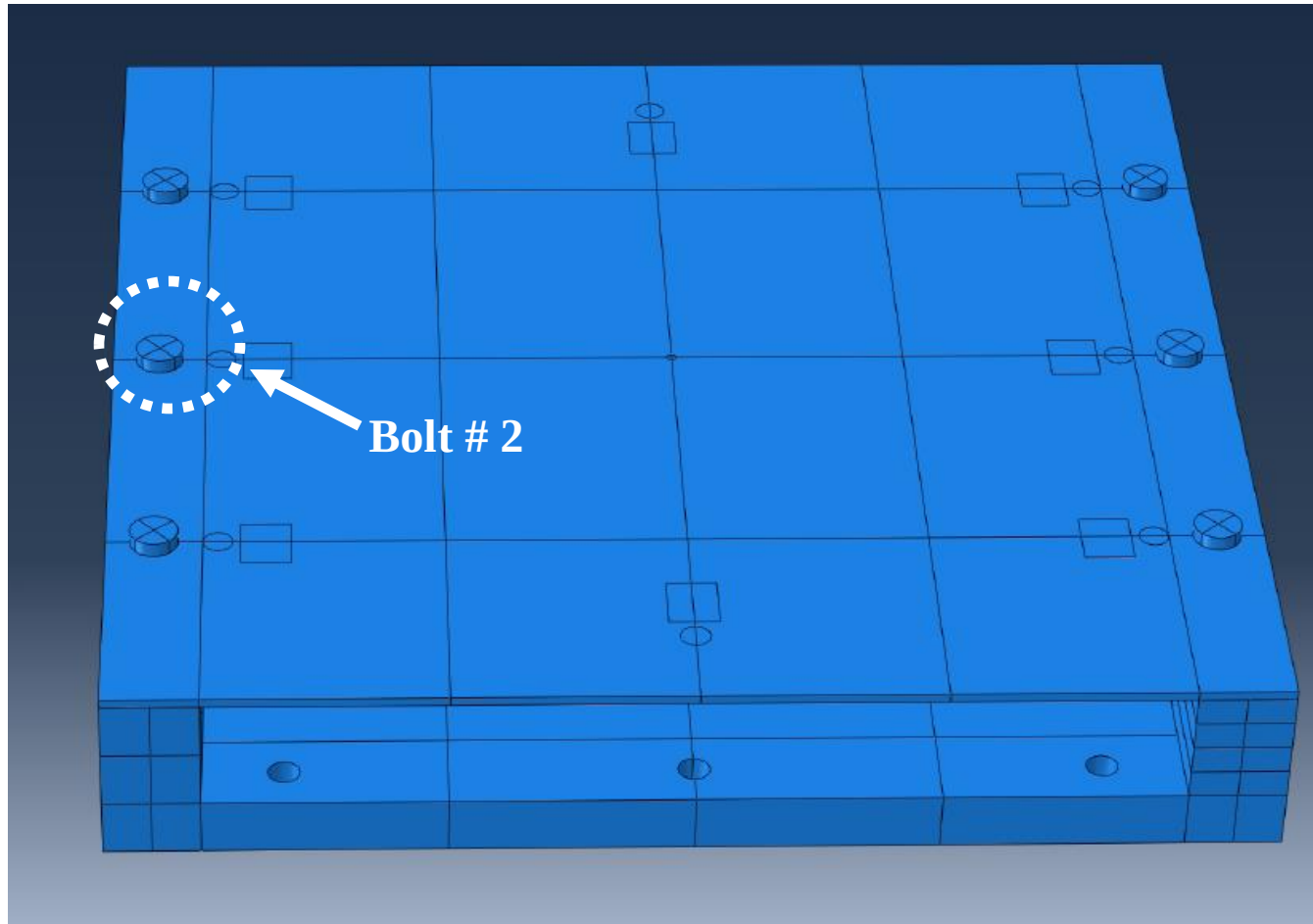
Structural Health Monitoring (SHM)

Self-powered, Wireless Sensors



Structural Health Monitoring (SHM)

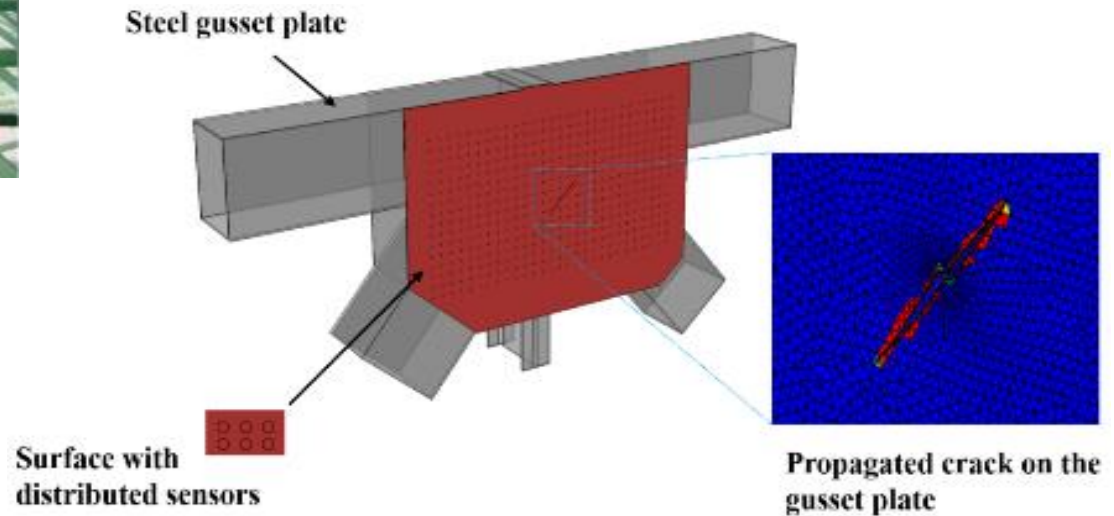
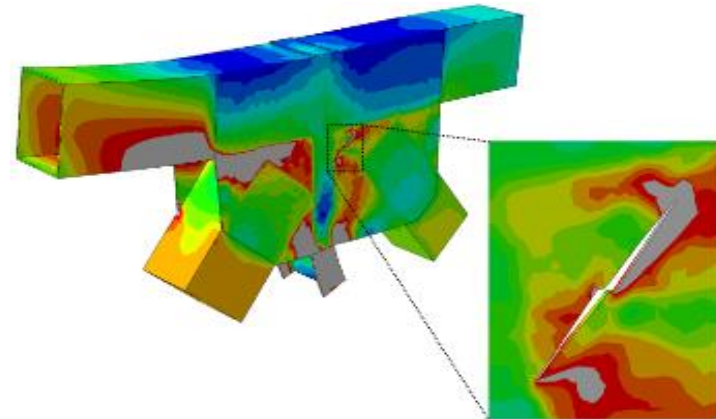
Bolted Steel Plates



Structural Health Monitoring (SHM)

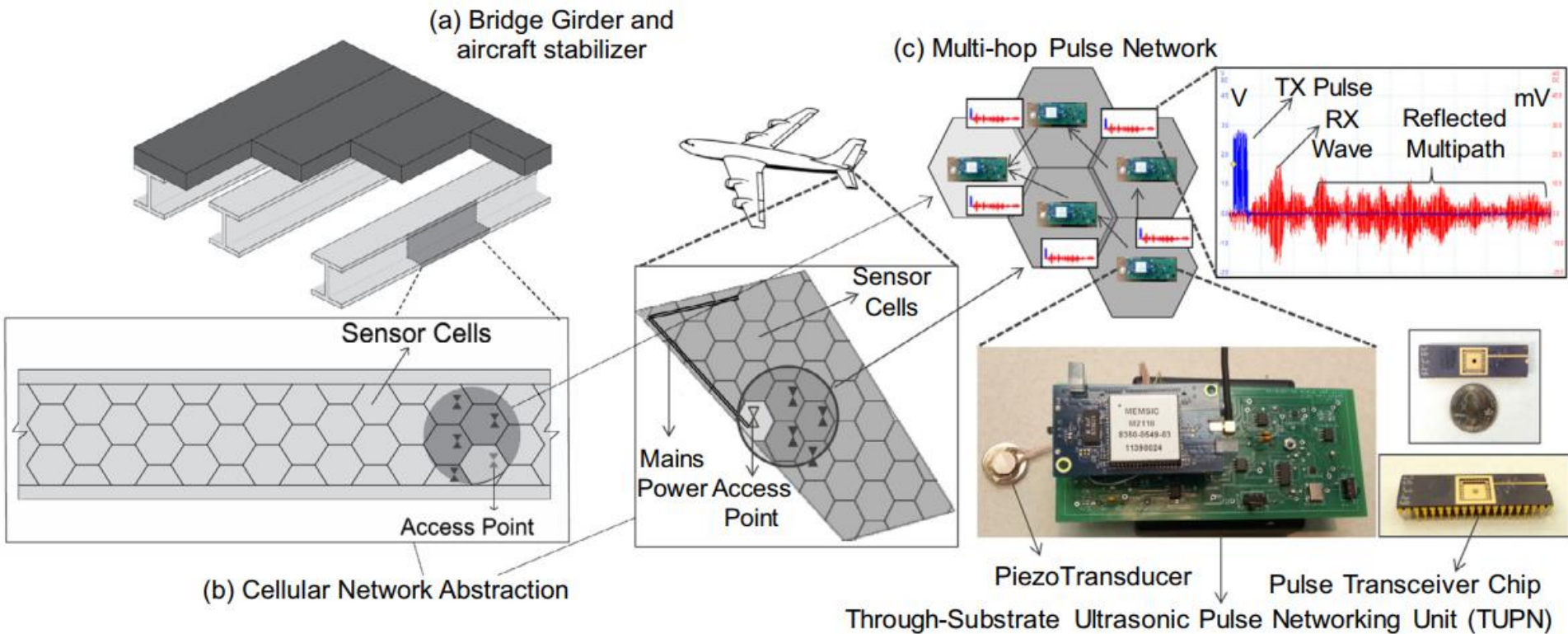
Gusset Plates

FE Simulations



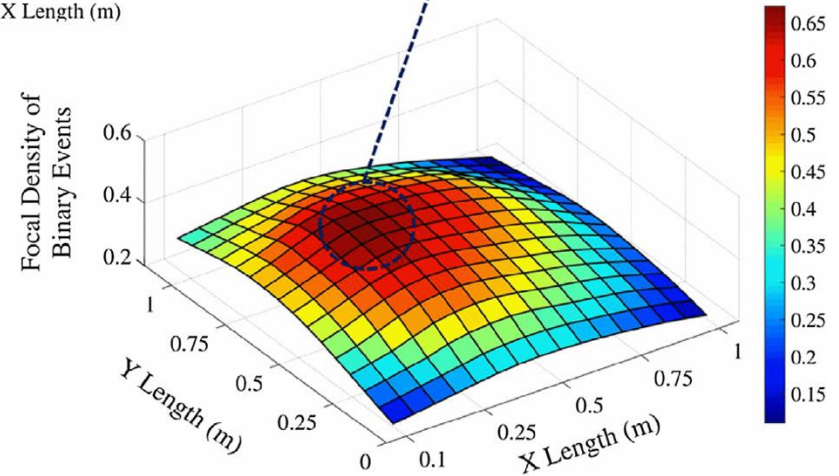
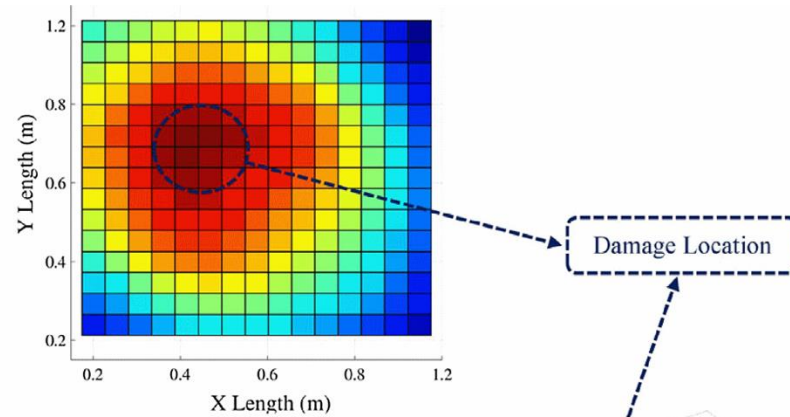
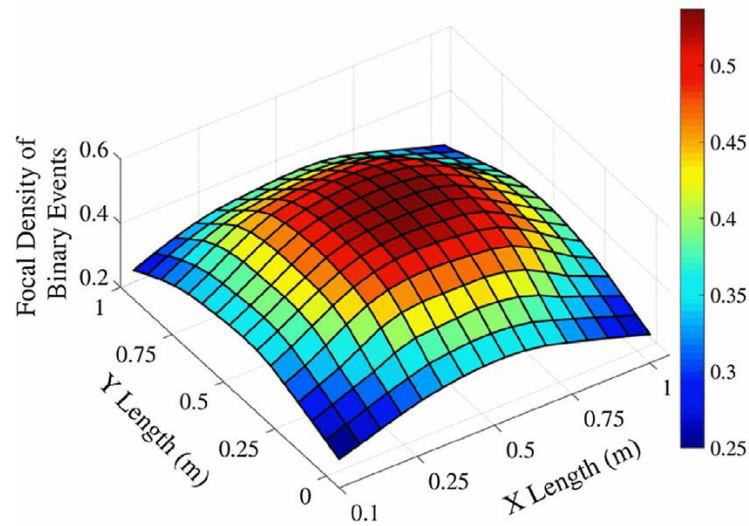
AI-Based Smart SHM

Piezo Sensing and PR



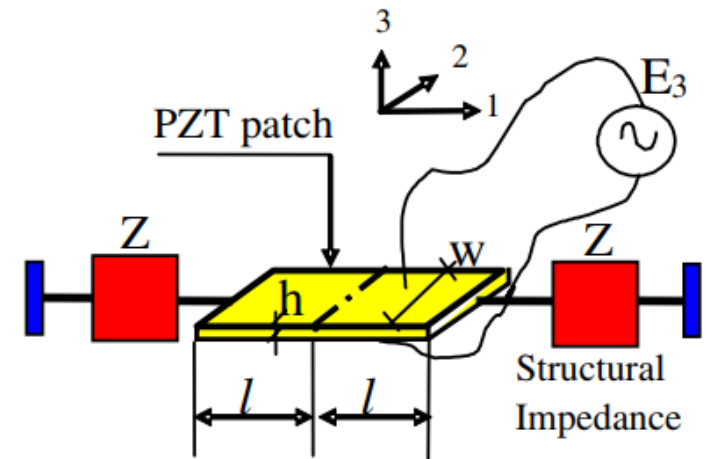
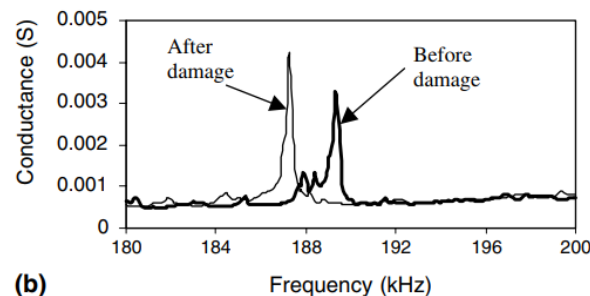
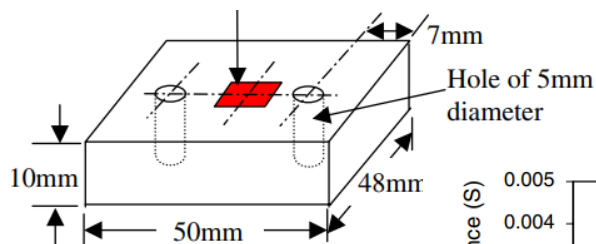
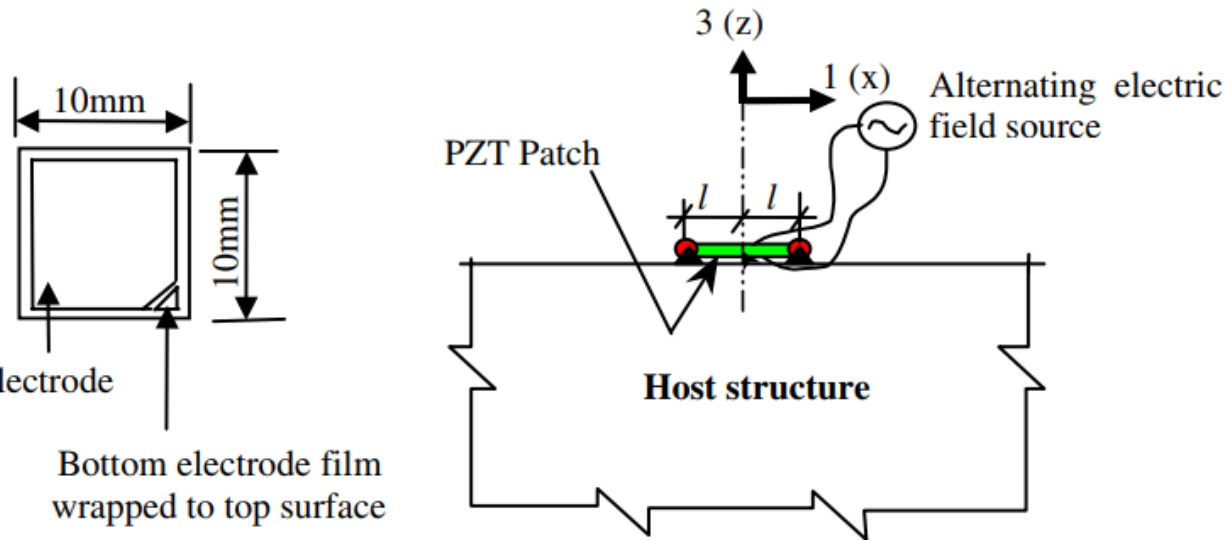
AI-Based Smart SHM

Piezo Sensing and PR



AI-Based Smart SHM

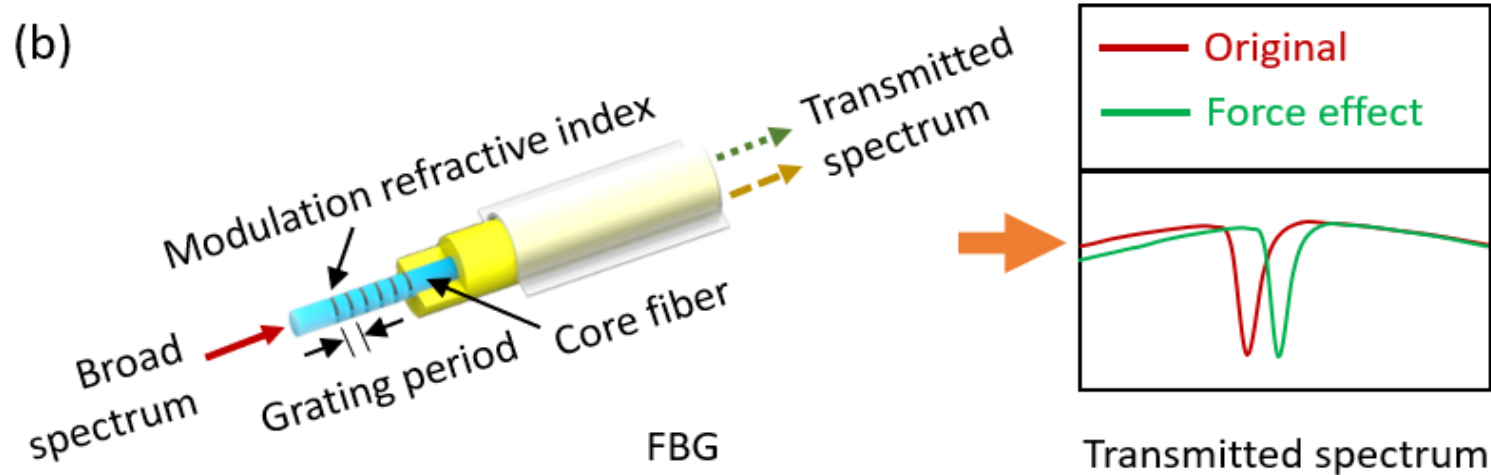
PZT Sensors



Electro-Mechanical Impedance (EMI)

AI-Based Smart SHM

Fiber Bragg Grating (FBG)



Bragg reflection wavelength of FBG

$$\begin{cases} \Delta\lambda_x = \lambda_x \left[-\frac{n_{\text{eff}}^2}{2E} \times \left[(p_{11} - 2\nu p_{12})\sigma_x + [(1 - \nu)p_{12} - \nu p_{11}] \times (\sigma_y + \sigma_z) \right] + \frac{1}{E} \times [\sigma_z - \nu(\sigma_x + \sigma_y)] \right] \\ \Delta\lambda_y = \lambda_y \left[-\frac{n_{\text{eff}}^2}{2E} \times \left[(p_{11} - 2\nu p_{12})\sigma_y + [(1 - \nu)p_{12} - \nu p_{11}] \times (\sigma_x + \sigma_z) \right] + \frac{1}{E} \times [\sigma_z - \nu(\sigma_x + \sigma_y)] \right] \end{cases}$$

AI-Based Metastructures

Quantitative Structure-Activity Relationships (QSARs)

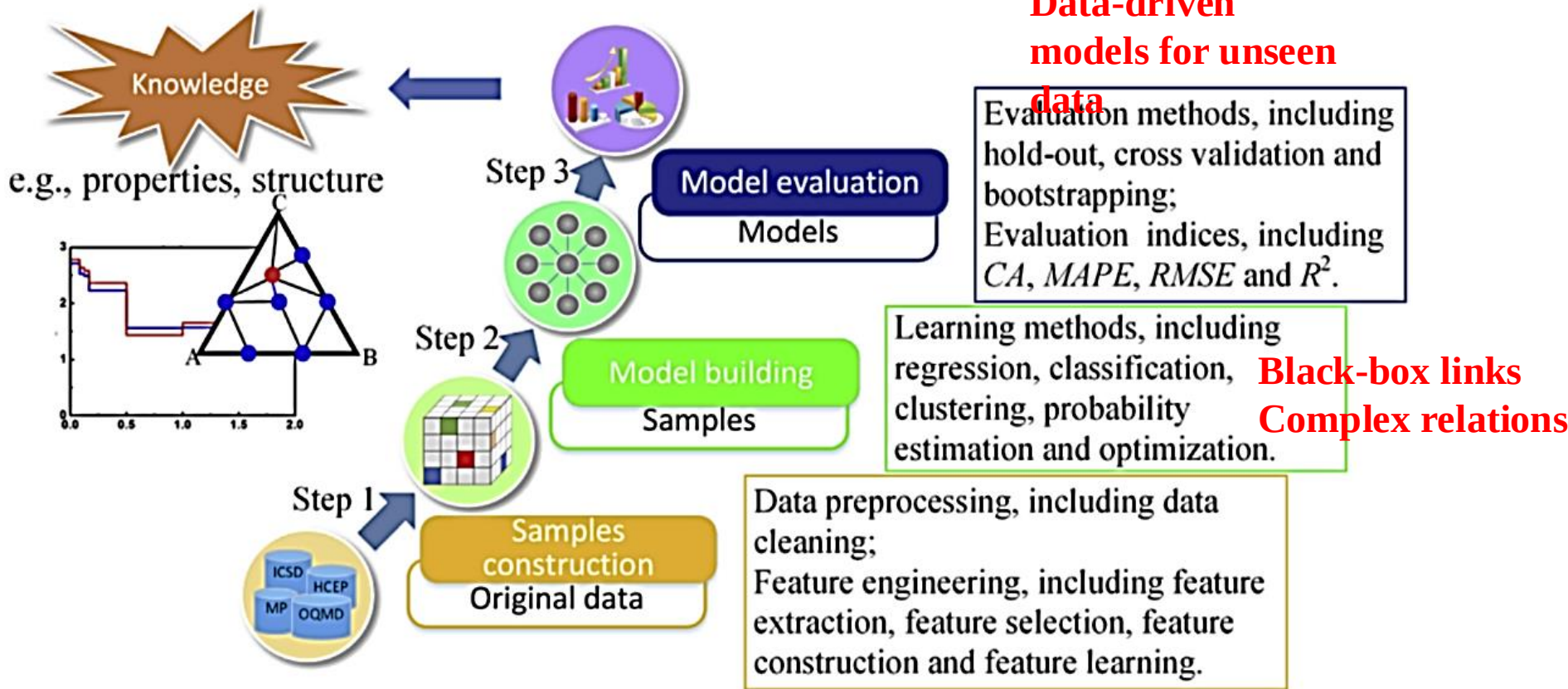


Goal + Sample + Algorithm = Model

- Goal: Given problem – Objective function
- Sample: Subset of population selected in prescribed manner
– Data preprocessing
- Algorithm: Self-contained step-by-step set of operations - ML (SVM, DT, ANN) and model optimization (GAs, SAs, PSO)

AI-Based Metastructures

Quantitative Structure-Activity Relationships (QSARs)



ML: Using examples of target functions (Complex relations) to find the coefficients with which a certain mapping function approximates the target function as closely as possible.

AI-Based Metastructures

The comparison of three evaluation methods.

Method	Advantages	Disadvantages
Hold-out	Low computational complexity.	The proper relative proportions of training/testing data are difficult to determine; The volume of the training data is smaller than that of the original dataset.
Cross-validation LOOCV	Not greatly influenced by changes in the volume of training data.	The computational complexity is high, especially on a large dataset; The volume of the training data is smaller than that of the original dataset.
Bootstrapping	Effective partitioning of training/testing data.	The distribution of the training data differs from that of the original dataset.

$$CA = S/N$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y'_i - y_i|}{y_i}$$

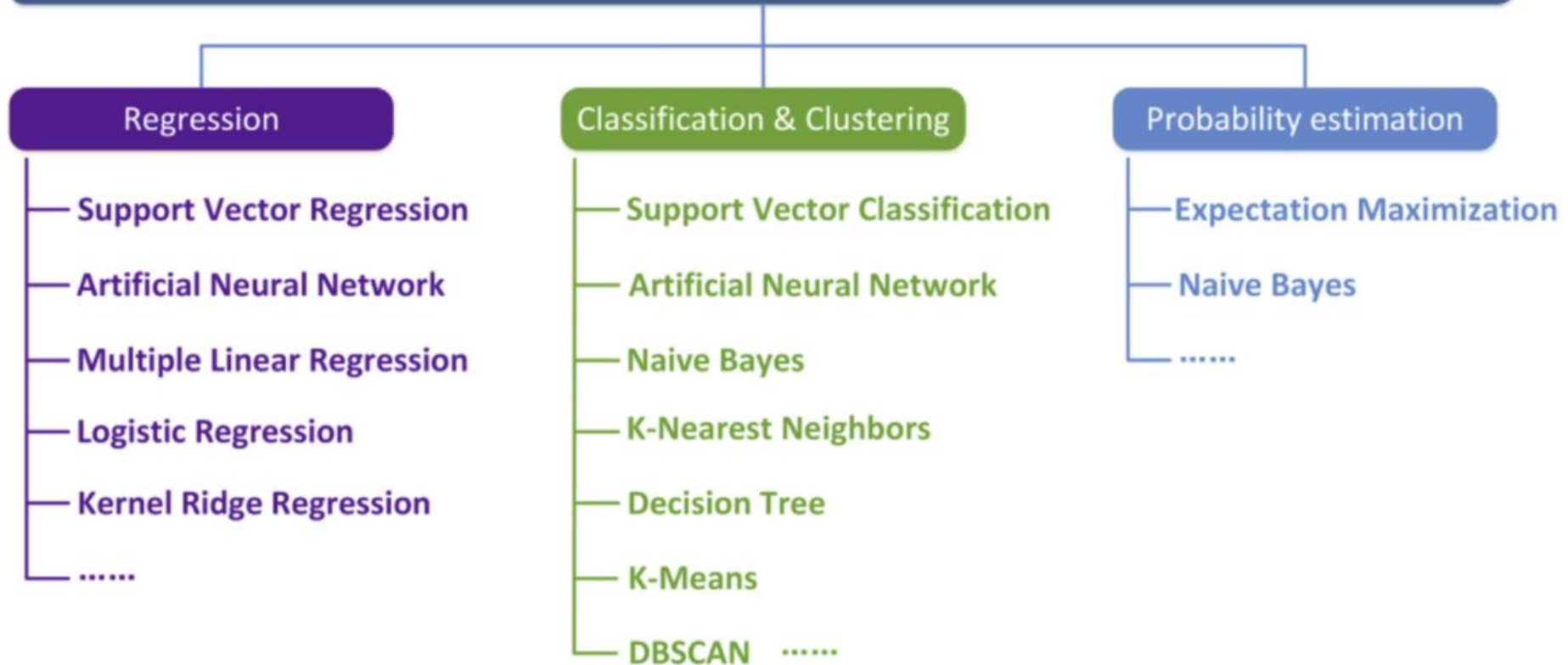
$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y'_i - y_i)^2}$$

$$R^2 = \frac{[\sum_{i=1}^n (y_i - \bar{y})(y'_i - \bar{y}')]^2}{\sum_{i=1}^n (y_i - \bar{y})^2 \cdot \sum_{i=1}^n (y'_i - \bar{y}')^2}$$

- Classification accuracy (CA)
- Mean Absolute Percent Error (MAPE)
- Root Mean Square Error (RMSE)
- Correlation Coefficient (R2)

AI-Based Metastructures

Commonly Used Machine Learning Algorithms in Materials Science

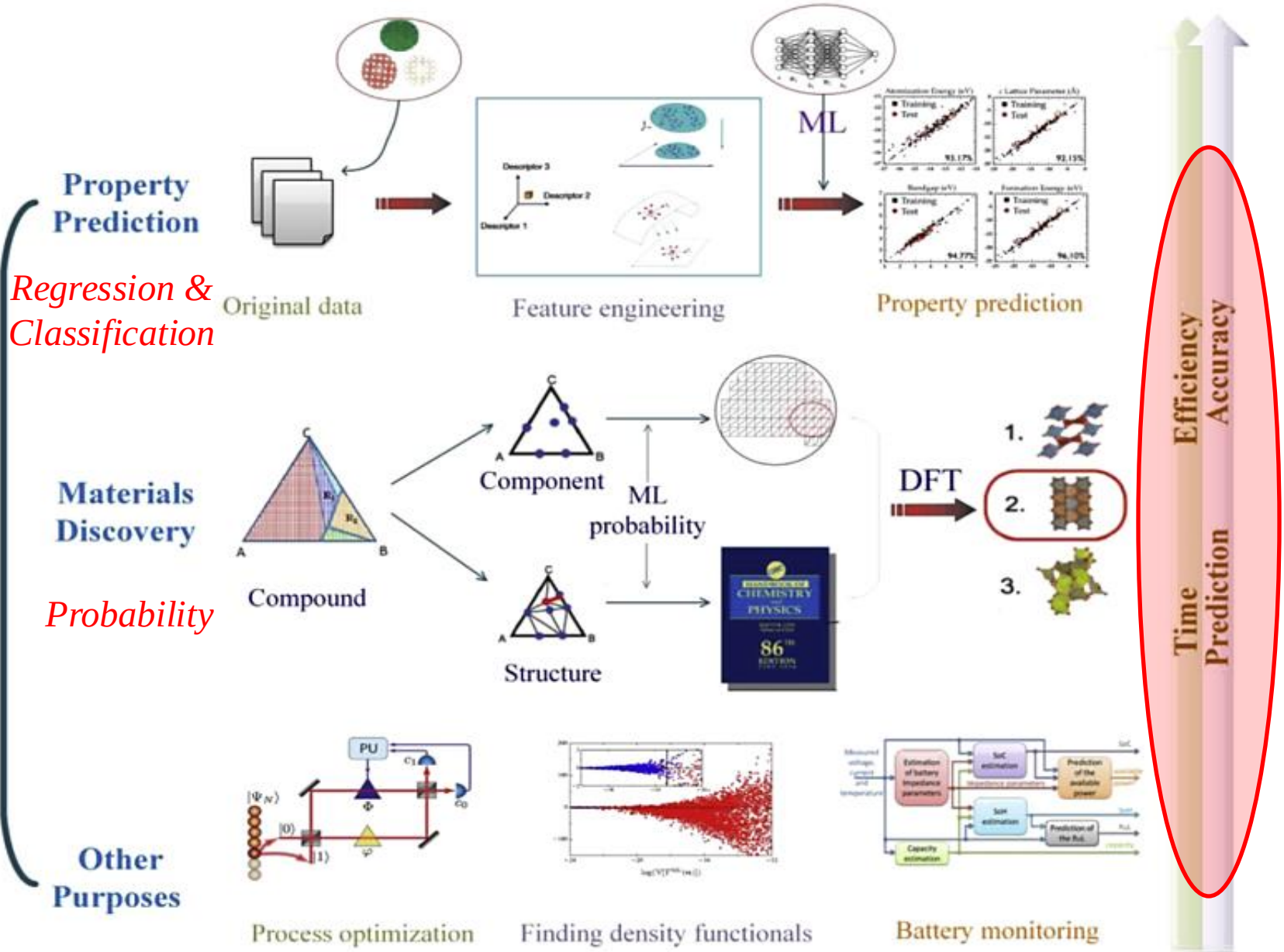


**Prediction of
Multiscale Structures**

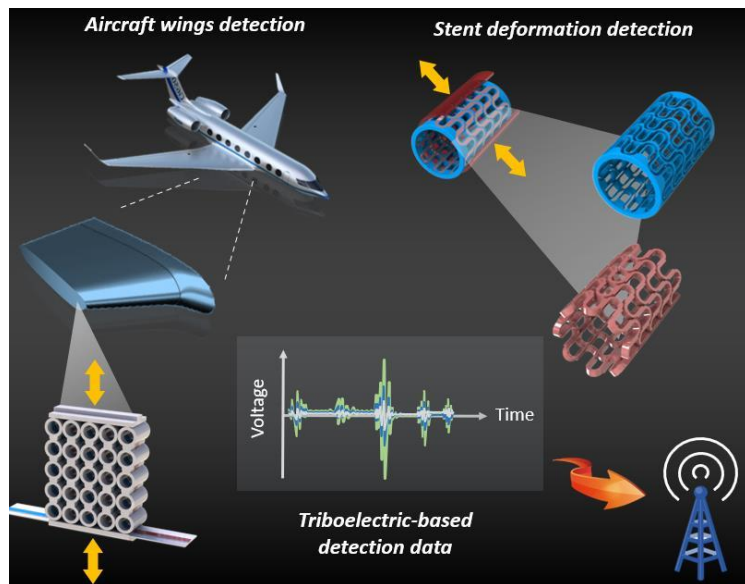
**Prediction of
Multiscale Structures**

New Structures

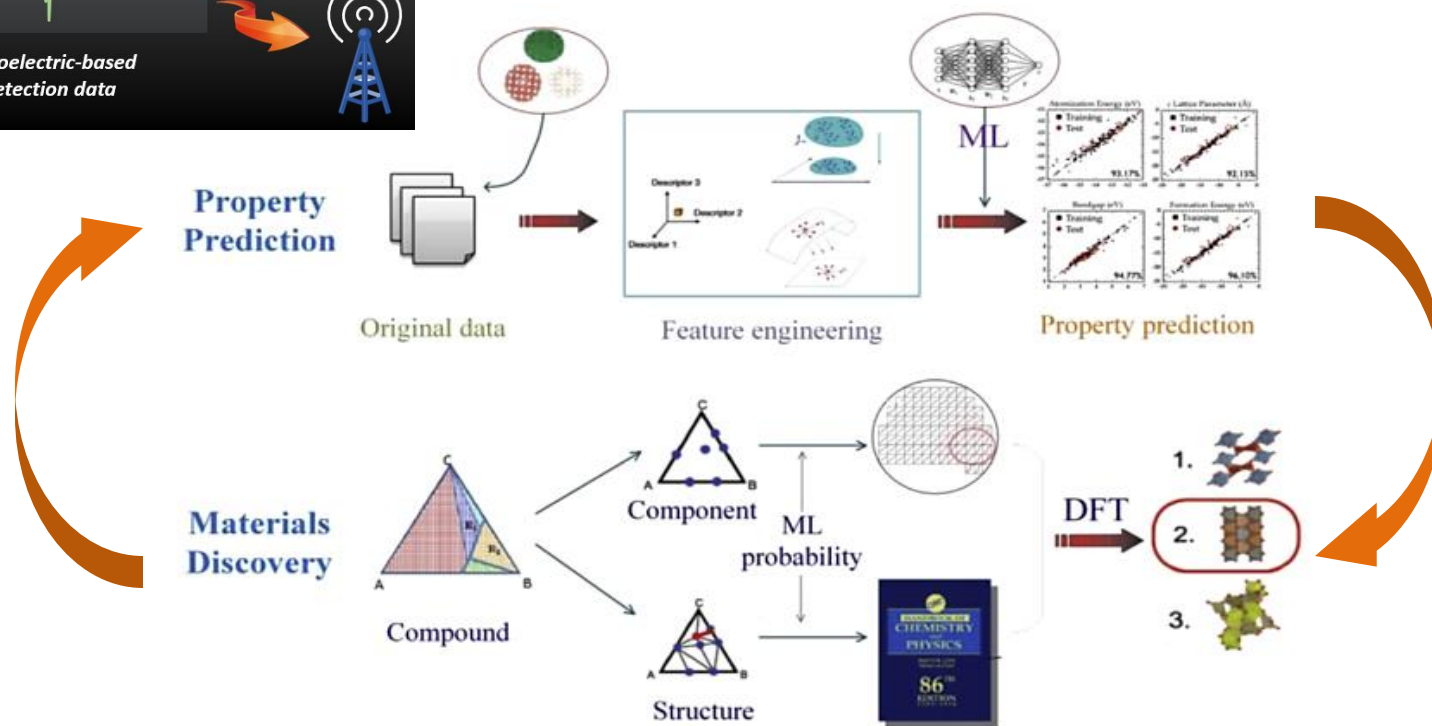
AI-Based Metastructures



AI-Based Metastructures



Design vs. Prediction

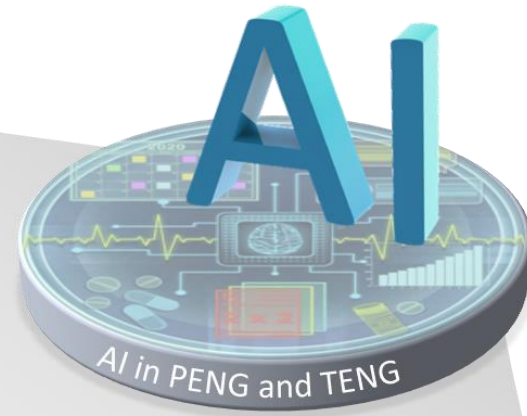


Generalized analysis theory



$$I_D = \varepsilon \frac{\partial E}{\partial t} + \left[\frac{\partial P_s}{\partial t} \right]$$

PENG TENG

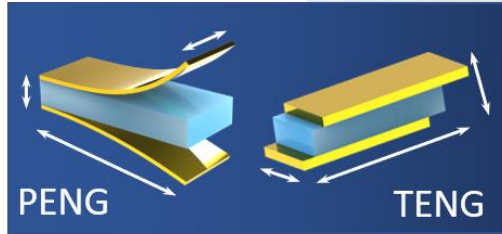


Analysis



Design

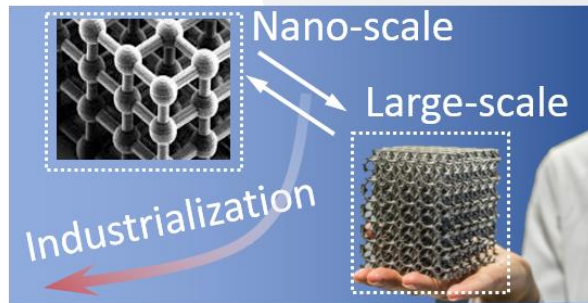
Advanced design strategies



Fabrication



Industrialized fabrication

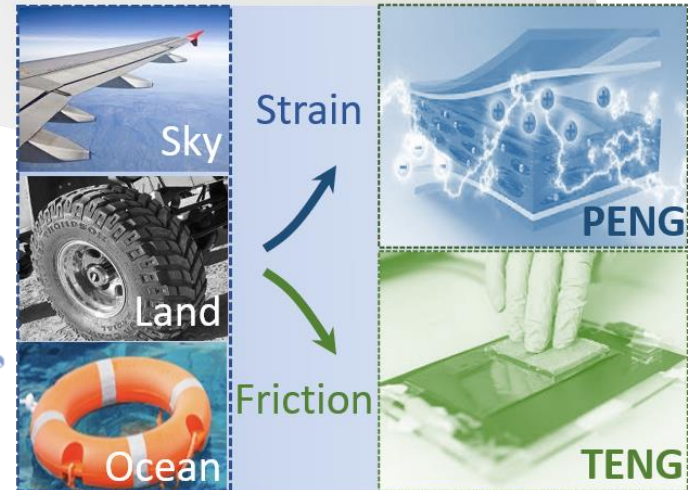


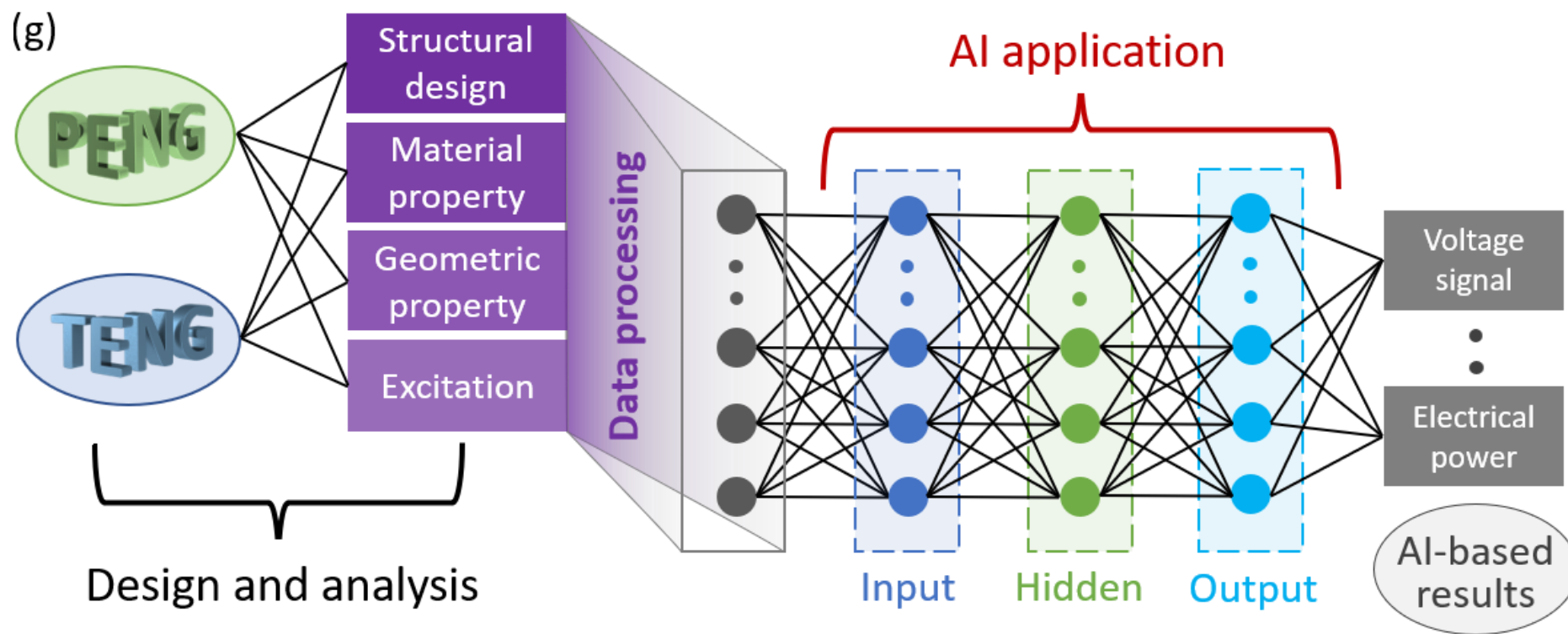
Application

AI for challenges in
PENG and TENG



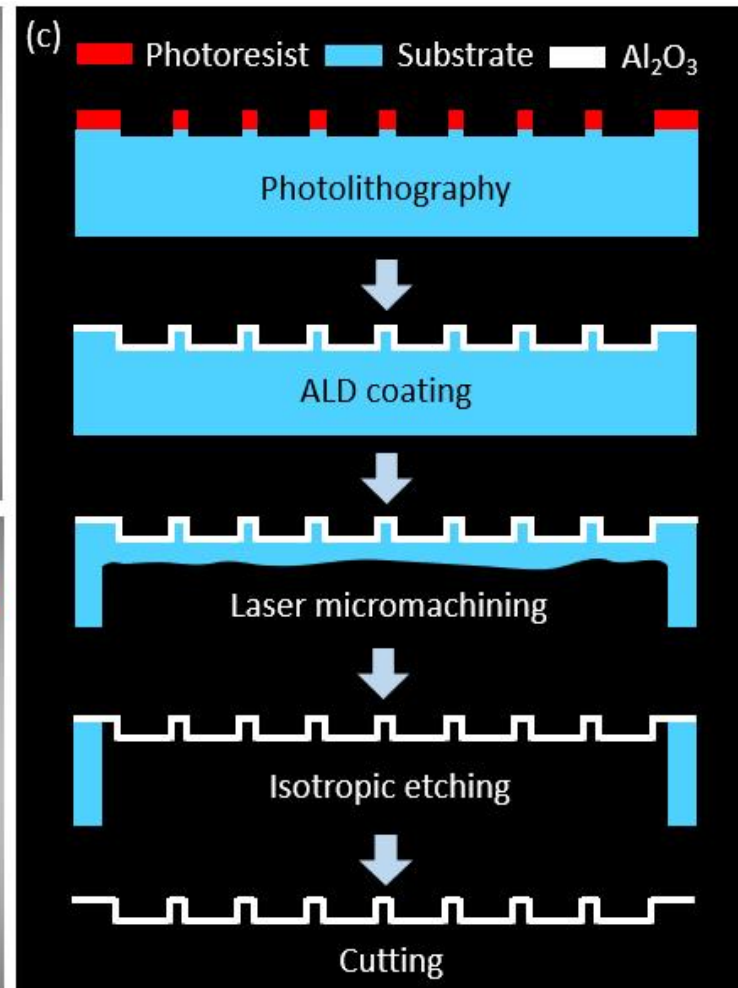
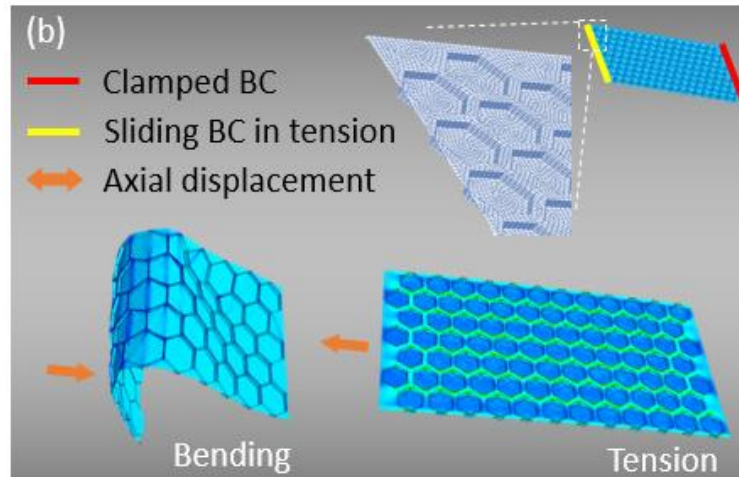
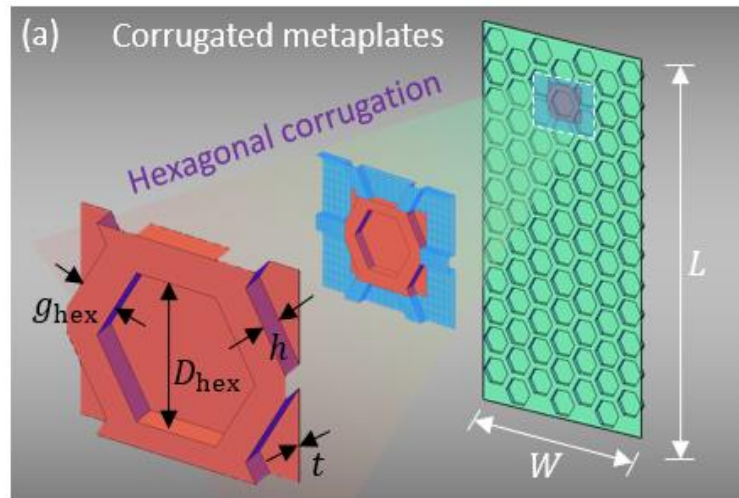
New function-enabled applications





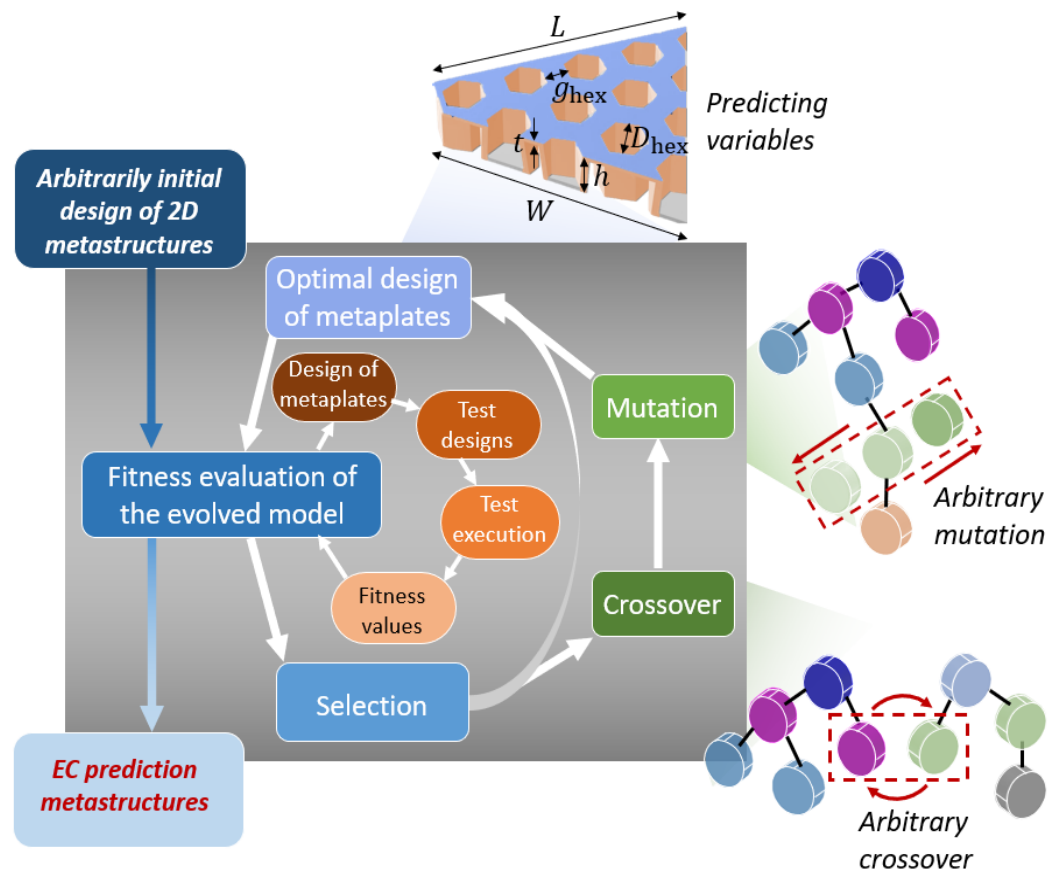
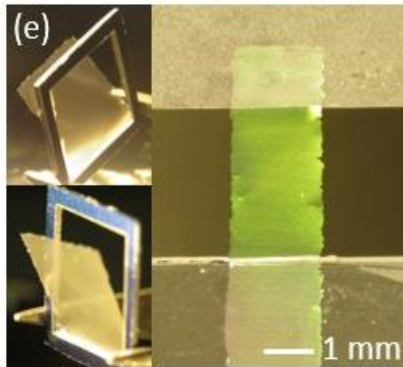
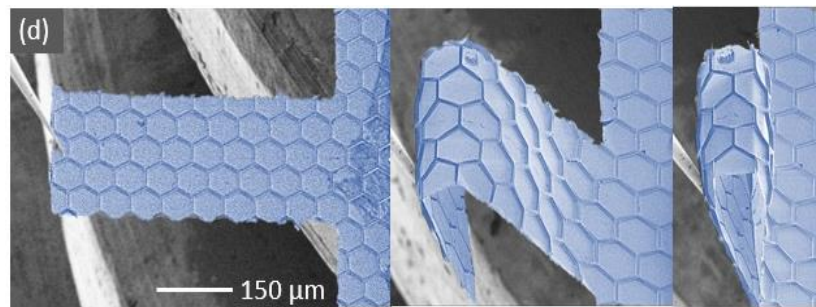
AI-Based Metastructures

Evolutionary Computation (EC)-Based Prediction



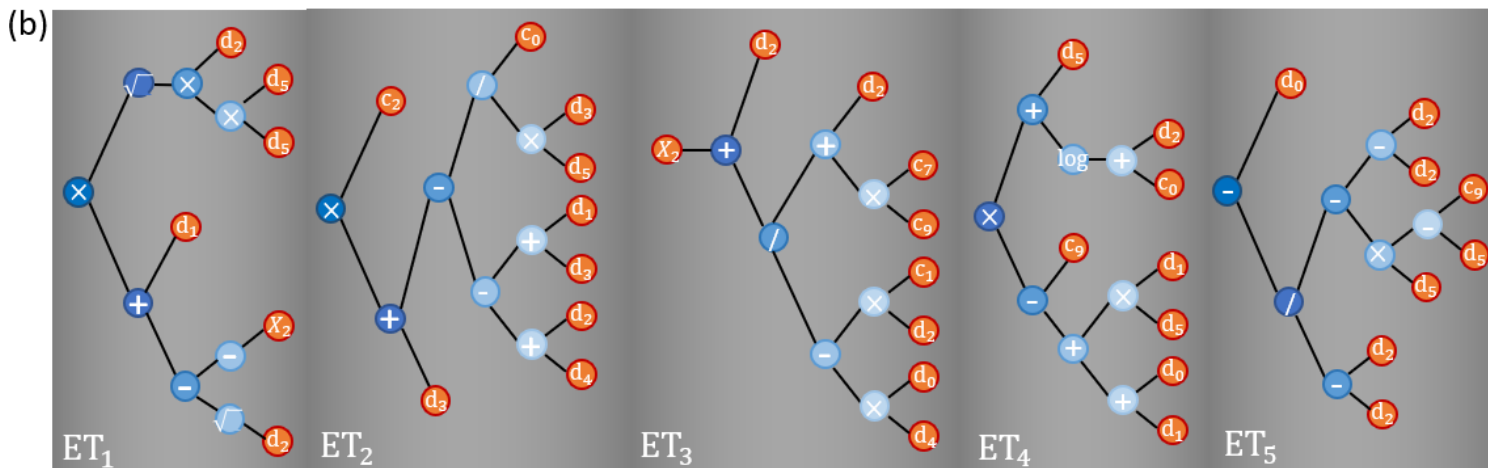
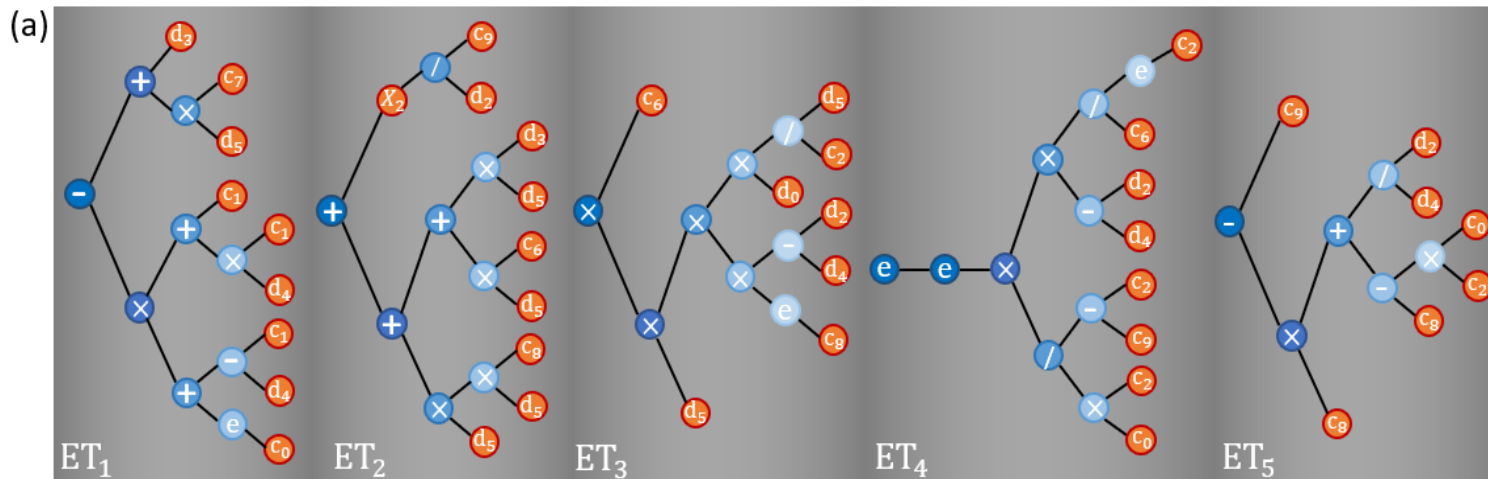
AI-Based Metastructures

Evolutionary Computation (EC)-Based Prediction



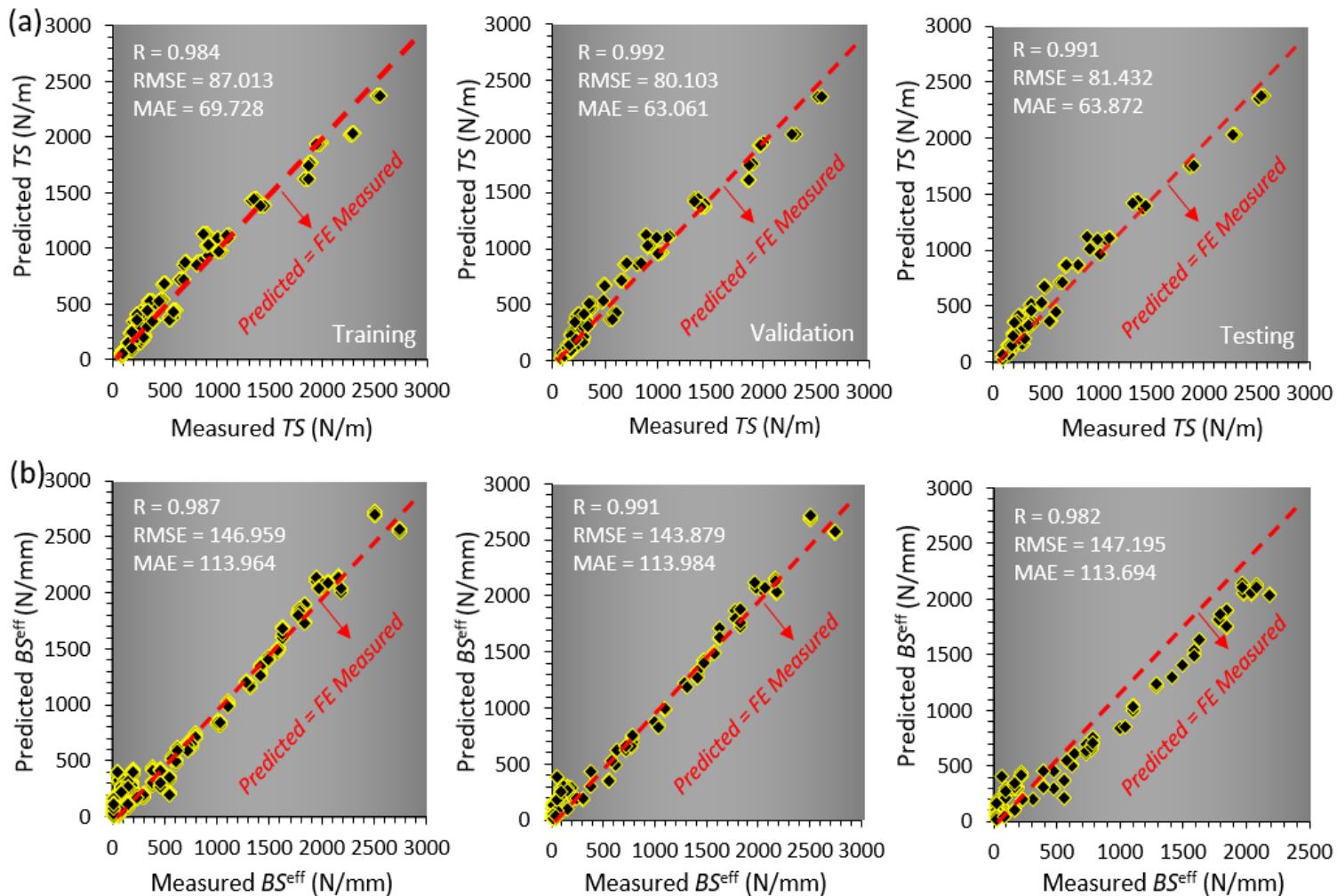
AI-Based Metastructures

Evolutionary Computation (EC)-Based Prediction



AI-Based Metastructures

Evolutionary Computation (EC)-Based Prediction



END

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